Feature-Band-Based Unsupervised Hyperspectral Underwater Target Detection Near the Coastline

Shuo Zhang, Puhong Duan, *Member, IEEE*, Xudong Kang, *Senior Member, IEEE*, Yan Mo, and Shutao Li, *Fellow, IEEE*

***Abstract*— With the improvement of imaging equipment, hyperspectral underwater target detection (HUTD) has raised much interest in recent years. The existing HUTD methods do not fully use spectral characteristics and need prior information about targets. Besides, the detection performance lacks verification in natural scenarios. In this article, the authors propose a feature-band-based unsupervised underwater target detection method (FBUD), which aims at finding the optimal feature bands to identify the underwater target near the coastline. Specifically, the normalized difference water index (NDWI) and unmixing technique are adopted to find the target and background pixels. Then, the spectral difference between the target and background is used to find the feature bands. With a simple and fast math operation of the feature bands, the probability map of the underwater target can be easily obtained. Besides, a new unmanned aerial vehicle (UAV)-borne hyperspectral image (HSI) dataset named HNU-UTD is built for underwater target detection in real-world scenes. Experimental results obtained with the HNU-UTD dataset confirm the accuracy and effectiveness of the proposed detection method, which even outperforms supervised detection methods.**

***Index Terms*— Feature band selection (BS), hyperspectral image (HSI), underwater target detection, unsupervised detection.**

I. Introduction

SPECTRAL characteristics and spatial distribution of different ground objects can be observed in hyperspectral images (HSIs). The abundant spectral information is helpful for distinguishing spatially similar objects which cannot be identified in natural images. Thus, target detection with HSIs has been extensively studied recently. Hyperspectral target detection (HTD) aims to find known targets which can be considered as a reconnaissance technique [1]. HTD applications include mineral exploration, environmental monitoring, toxin tracking, rescue, and search in disaster areas [1], [2], [3], [4].

Generally, the task of HTD is to detect targets of interest in the image with prior knowledge of the given target. A simple way to realize HTD is to measure the spectral distance between the known target pixels and other pixels. The commonly used similarity evaluated indexes are spectral angle mapping (SAM) and Euclidean distance (ED). This kind of methods is simple but easily affected by spectral variation and the quality of prior target spectral features [5]. The alternative methods are statistics-based methods, such as adaptive coherent/cosine estimator (ACE) [6], and matched filter (MF) [7]. The MF and ACE methods are developed under binary hypothesis testing and derived from the generalized likelihood ratio test (GLRT). Besides, the subspace-based HTD method is also widely investigated. Among them, the orthogonal subspace projection (OSP) [8] is a representative algorithm that projects the pixels to an orthogonal subspace where the background can be suppressed and the target signature signal-to-noise ratio (SNR) can be improved. However, these traditional detection methods usually rely on the hypothesis model and statistical distribution which are not reasonable in real-world applications. They may fail when the assumptions cannot be met. Other HTD methods without assumption on the background statistics are also developed, such as constrained energy minimization (CEM) [9] and sparse-representation-based methods [10], [11], [12]. CEM uses a finite impulse filter to minimize the output energy while the response of the target signature is constant. Based on CEM, a series of improved methods are proposed [13], [14], [15], [16]. For example, Zou and Shi [14] propose a hierarchical CEM (hCEM) which uses a layer-by-layer filtering procedure to separate the target and the background. Jiao and Chang [15] extend the CEM to its kernel versions, which adopts various kernels to obtain a higher dimensional feature space where the CEM can be performed. Sparse-representation-based methods constructed an overcomplete dictionary to reconstruct test pixels and detect targets with the reconstruction error. For instance, sparse and collaborative representations are combined for HTD in [17]. To make full use of spatial information, Imani [18] uses attribute profiles to extract spatial features. A spatially

Manuscript received 22 November 2022; revised 28 February 2023; accepted 30 March 2023. Date of publication 20 April 2023; date of current version 11 May 2023. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFA0715203; in part by the National Natural Science Foundation of China under Grant 62221002, Grant 61890962, Grant 61871179, and Grant 62201207; in part by the National Science Foundation of Hunan Province under Grant 2020GK2038; in part by the Hunan Provincial Natural Science Foundation for Distinguished Young Scholars under Grant 2021JJ022; in part by the Huxiang Young Talents Science and Technology Innovation Program under Grant 2020RC3013; in part by the Defense Industrial Technology Development Program under Grant JCKYS2021604SSJS010 and Grant YGYGBYY-9JY052; and in part by the Fellowship of China Postdoctoral Science Foundation under Grant 2022M721106. *(Corresponding author: Xudong Kang.)*

Shuo Zhang, Puhong Duan, and Shutao Li are with the College of Electrical and Information Engineering, Hunan University, Changsha 410082, China (e-mail: z_shuo@hnu.edu.cn; puhong_duan@hnu.edu.cn; shutao_li@hnu.edu.cn).

Xudong Kang is with the School of Robotics, Hunan University, Changsha 410082, China (e-mail: xudong_kang@163.com).

Yan Mo is with the College of Electrical and Information Engineering, Changsha University of Science and Technology, Changsha 410082, China (e-mail: moyan@csust.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3268299

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:03:13 UTC from IEEE Xplore. Restrictions apply.

adaptive sparse representation HTD method is also developed in [19].

Traditional HTD methods only can extract shallow and linear features, which are limited in complex scenes. Due to the capability in representing high-level features, deep learning (DL)-based methods have been widely used in HSI processing. DL-based HTD methods also have attracted extensive attention recently [20], [21], [22], [23]. Since the target samples are usually insufficient for DL-based HTD tasks, many researchers focus on the small sample size problem. For example, Zhang et al. [24] use a modified autoencoder and linear prediction strategy to enlarge the training sample set. Then, the generated pixel-pairs are used as the input of convolutional neural network (CNN) to learn the discriminative similarity and yield the output. Zhu et al. [25] adopt a hybrid sparse representation and pixel selection strategy to find enough background samples. Target samples are generated with target prior spectrum and background samples. Then, a two-stream CNN is designed to detect the target. In addition, the limited spatial resolution of HSI and diversity of ground objects result in mixed pixels. The subpixel target can be considered as a mixed pixel composed of target and background spectrum. Thus, the spectral difference between the target and background is small, which is a challenge in HTD [26], [27]. To solve this problem, Huang et al. [28] extract subpixel, pixel, and superpixel features and then fuse the multiscale features to separate targets. Rao et al. [29] generate the subpixel target and mixed background samples for a Siamese transformer network which can extract global features with long-range dependencies to achieve HTD. Vincent and Besson [30] derive the exact GLRT for subpixel target detection. This detector can be used with a local and a global covariance estimation window, which is superior to the standard target detectors on small covariance estimation windows.

Although various HTD methods have been developed, the majority of them focus on detecting land-based targets by default. When the targets of interest are underwater, the assumption that the targets are essentially independent of the background is not valid in many land-based HTD methods. In the hyperspectral underwater target detection (HUTD) task, the spectrum of target is not stable with the influence of water environment. In other words, the water-leaving radiation is associated with the inherent optical properties (IOPs) of the water. To achieve HUTD, Jay et al. [31], [32] insert a bathymetric model to correct the spectral distortions introduced by water. Then, traditional land-based algorithms (i.e., MF, adaptive MF, ACE) are used to detect underwater targets. Gillis [33] uses physical and nonlinear mathematical models to estimate the underwater target space in a specified scene. Then, a nonlinear mathematical technique, manifold learning, is used to exploit the geometrical structure in the target space to reduce dimensionality and simplify the HUTD problem. Furthermore, Xia and Gu [34] develop a parameter feature extraction method for HUTD. Specifically, the features are considered as the inversed water depth and bottom reference reflectance which are obtained by a quasi-analytical model. This model can inverse the characteristics of the seawater environment according to the radiated spectrum.

However, the performances of the above methods rely on the accuracies of IOPs estimating results. To get accurate IOPs, enough prior information about the water environment is needed. Unfortunately, the characteristics of the water environment are always unknown. The IOPs of water are also difficult to be estimated precisely in practice. In this situation, a GLRT-based bathymetric filter is also proposed to estimate unknown parameters and detect targets at the same time in [32]. But GLRT assumes that unknown parameters follow a multivariate normal distribution, which may not fully describe the correlation among different unknown parameters and have an adverse impact on the effectiveness of HUTD results. Besides, Qi et al. [35] take the depth information into account and design a UTD-Net that combines a bathymetric model with an autoencoder. The output of UTD-Net is the target-associated abundance maps that be fused to detect targets. Qi et al. [36] also propose an iterative HUTD framework named SUTDF to make use of the depth information. SUTDF uses spectral–spatial contextual information to generate a guidance dataset for depth information estimation. The depth information is then used to produce a more diversified dataset for the target detection and improve the detection performance. At the same time, the detection result will be used to enrich the guidance dataset for depth estimation. Through the updating process, the detection result of the network is more accurate. In addition, some DL underwater target detection methods using other images as input have been developed recently [37], [38], [39]. For example, a lightweight detection network with underwater RGB images is proposed in [37], which can well solve the problem of underwater image color distortion. The fine-tuned YOLOv2 and YOLOv3 models are used to detect marine species in [38].

From the perspective of improving efficiency and eliminating redundant spectral information, band selection (BS) is also combined with the target detector for HUTD [40], [41]. Fu et al. [40] propose a constrained-target optimal index factor to select band subsets related to the targets, and then use CEM to find the desired underwater targets. In [41], a bathymetric model is embedded into the BS progress. The authors develop a bathymetric latent spectral representation learning scheme to construct a subspace which is used as the reference to pick out desired bands. Then, an iteration strategy is used to generate a diversified band subset but the iteration operation brings the time computational burden.

Overall, only a few studies focus on HUTD in the literature at the present time. In another word, HUTD has not been explored extensively within the field of HSI analysis. There are some issues that still need to be solved. First, there is no public dataset for HUTD. Though the above methods have achieved great success in HUTD, they mainly tested on simulation datasets and do not make the data publicly available. The detection performance in real-world scenes is still not verified. Second, the majority of methods are supervised and depend on the prior information of the target and water environment, which limits the applicability of these methods. Furthermore, these methods consider a lot of information, such as depth and IOPs. These information are helpful but not able to be estimated accurately. Considering the above-mentioned issues,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:03:13 UTC from IEEE Xplore. Restrictions apply.

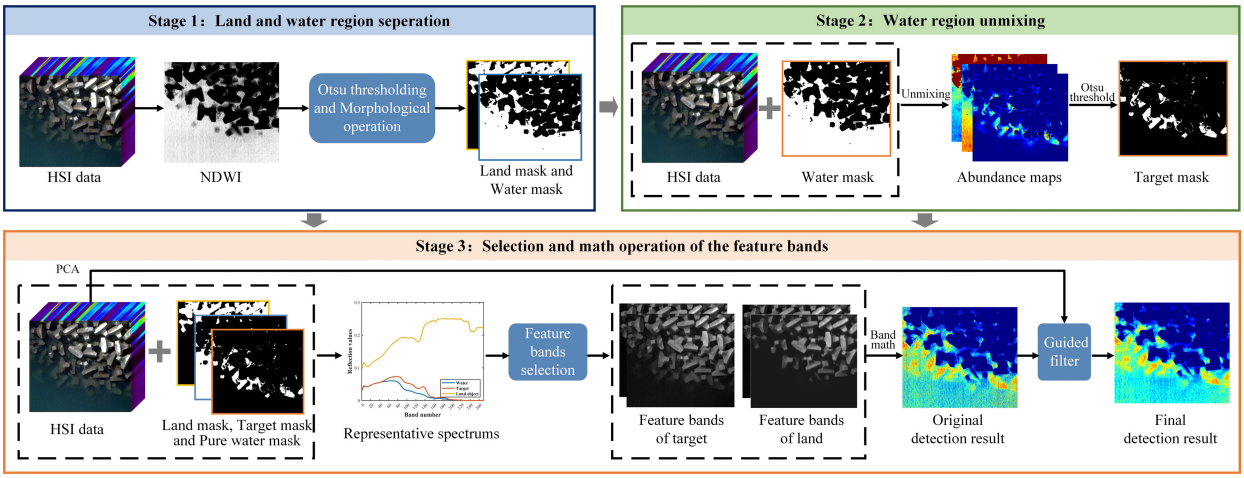


Fig. 1. Flowchart of the proposed FBUD framework.

we try to find a simple but effective way to achieve underwater target detection in real-world scenes. Concretely, a feature-band-based unsupervised underwater target detection method (FBUD) method is proposed for detecting underwater targets near the coastline in this article. The feature bands are selected by comparing the spectrums of different objects extracted with an unmixing technique. Then, band math is performed on the feature bands to obtain the probability distribution map of the target. The contributions of this article are summarized as follows.

1) An FBUD is proposed for HUTD. The proposed method leverages abundance maps to find representative pixels for feature BS, which can tackle the sample size problem.
2) An open-source unmanned aerial vehicle (UAV)-borne dataset named HNU-UTD[1] is constructed for underwater detection for the first time. The HNU-UTD dataset fills the gaps of the dataset missing in HUTD.
3) Experiments carried out on the HNU-UTD dataset confirm that the proposed unsupervised method can achieve comparable results with supervised detection approaches.

II. Proposed Method

In this section, the proposed FBUD method is introduced in detail. The purpose of FBUD is to find the feature bands which can separate the target and background effectively. Fig. 1 presents the flowchart of FBUD, which mainly consists of three parts: land and water region separation, water region unmixing, and feature BS and operation. The procedure of these steps is detailed as follows.

A. Land and Water Region Separation

To separate land and water region, the normalized difference water index (NDWI) [42] is used to extract water regions using the water spectral reflectance in green (G) and near-infrared radiation (NIR) bands. Given an HSI $\mathbf{I} \in \mathbb{R}^{M\times N\times D}$ with a known wavelength range, the NDWI is calculated as follows:

$$\text{NDWI} = \frac{\mathbf{I}_{\text{G}} - \mathbf{I}_{\text{NIR}}}{\mathbf{I}_{\text{G}} + \mathbf{I}_{\text{NIR}}}. \tag{1}$$

[1]https://github.com/lotuszs/HNU-UTD-dataset

Then, a threshold of experience is used for NDWI to get the water mask $\mathbf{M}_W$ and land mask $\mathbf{M}_L$ by the following equations.

$$\mathbf{M}_W^* = \begin{cases} 1, & \text{if NDWI} > 0 \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

$$\mathbf{M}_W = \mathbf{M}_W^* \cdot W_{\text{se}}$$
$$\mathbf{M}_L^* = 1 - \mathbf{M}_W^*$$
$$\mathbf{M}_L = \mathbf{M}_L^* \circ W_{\text{se}} \tag{3}$$

where $\circ$ and $\cdot$ denote the morphological opening and closing operation, respectively. W_{se} is a flat disk-shaped structuring element with a three pixels' radius. The preliminary water mask $\mathbf{M}_W^*$ and preliminary land mask $\mathbf{M}_L^*$ are obtained through binary processing NDWI, which are not accurate on the boundary. Therefore, the morphological operation is adopted to reduce wrong classified pixels.

B. Water Region Unmixing

The motivation of this step is to obtain the target map with an unmixing model in an unsupervised way. Considering the diversity of land objects, it is necessary to eliminate the influence of land objects on the endmember extracting process. Thus, the endmember extracting method [43] and unmixing model [44] are adopted to unmix the water region of an image with the water mask $\mathbf{M_W}$.

The endmember extracting method [43] assumes that the volume contained by a simplex formed of the purest pixels is the largest in D spectral dimensions. At first, a set of pixels are randomly selected as endmembers. Then, the volume is calculated for each pixel and each endmember. The endmember is replaced with the pixel and the volume is recalculated. If the recalculated volume is larger, the endmember is updated with the corresponding pixel. This procedure is repeated until the volume reaches the max value.

To estimate the volume, the dimensionality of $\mathbf{I}$ is reduced to be one less than the number of endmembers with the maximum noise fraction transform. In this work, the number of endmembers l is set as three, since the land, target, and pure water are considered. The volume of the simplex formed by the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:03:13 UTC from IEEE Xplore. Restrictions apply.

endmember is proportional to the matrix of $\mathbf{E}$, the endmembers augmented with a row of ones. The volume $\boldsymbol{V}$ and matrix $\mathbf{E}$ can be expressed as

$$\mathbf{E} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ e_1 & e_2 & \cdots & e_l \end{bmatrix} \tag{4}$$

$$\boldsymbol{V}(\mathbf{E}) = \frac{1}{(l-1)!}\text{abs}(|\mathbf{E}|) \tag{5}$$

where e_i is the ith endmember, and $l-1$ is the number of dimensions occupied by the data.

Once endmembers are found, all the pixels in the HSI can be unmixed to obtain the abundance map for each endmember. In general, the linear mixing model with the endmembers is used to represent each pixel

$$x = \sum_{i=1}^{l} a_i e_i + \varepsilon \tag{6}$$

where x is the $D \times 1$ received pixel spectrum, and a_i is the mixing proportion of the ith endmember. ε is a Gaussian random error. Then, the endmember contributions of pixel x can be determined by least squares inversion as follows [44]:

$$\mathbf{A} = \left(\mathbf{E^T E}\right)^{-1}\mathbf{E^T} x. \tag{7}$$

According to the unmixing result, an adaptive binary operation [45] is performed on the abundance map of underwater target $\mathbf{AM}_T$. Since there is no prior information of the target and background, the object represented by each abundance map is unknown. Thus, the characteristic of water that the reflectance value in the NIR band is very lower is used to identify $\mathbf{AM}_T$. Specifically, two rules are used to identify $\mathbf{AM}_T$: First, the land mask region $\mathbf{M}_L$ in the abundance map $\mathbf{AM}_T$ should have a lower value. Second, the spectral reflectance of the NIR band corresponding to the binarized $\mathbf{AM}_T$ (target map) is not the lowest.

The procedure of obtaining target map $\mathbf{M}_T$ can be represented by $\mathbf{M}_T = \text{Otsu}(\mathbf{AM}_T)$, where $\text{Otsu}(\cdot)$ denotes the Otsu threshold method [45]. After that, the pure water map $\mathbf{M}_{\text{PW}}$ is equal to $\mathbf{M}_W - \mathbf{M}_T$.

C. Selection and Math Operation of the Feature Bands

In this procedure, the spectral difference in different objects is used to select feature bands that can highlight changes in underwater target and suppress changes in the background at the same time. Specifically, two groups of feature bands are selected, in which one is used to detect underwater target and the other is used to detect land regions. The land region detection result is used to optimize the underwater target detection result. The specific procedure of feature BS is described below.

1) *Generating the Representative Spectrums of Different Objects:* The representative spectrums of the land region, pure water, and target are considered as the mean spectrum of their pixels, respectively, which can be obtained by the corresponding masks. Assume the representative spectrums of land region, pure water, and target are s^l, s^w and s^t, respectively. s^w is regarded as the mean spectrum of land region, $s^l = \text{mean}(\mathbf{I}(\mathbf{M}_L == 1, :))$. s^w and s^t can be calculated in the same way. Besides, the mean filter is applied to these representative spectrums to reduce noise interference.
2) *Selecting the Feature Bands of Underwater Target:* The bands meeting the requirements of a large difference between target and pure water and a small difference between land objects are considered as feature bands. The details are summarized as follows.
   a) Calculating the difference spectrum of s^t and s^w, s^{tw} ($s^{tw} = s^t - s^w$). Then, finding the peak and valley of s^{tw}, and the corresponding bands are labeled as $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$, respectively. $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ represent the image bands with the largest and smallest difference between the target and pure water, respectively.
   b) Finding the stable band ranges $\mathbf{br}_i, i \in (1, \ldots, n)$ where the values of s^{tw} are similar. $\mathbf{br}_i$ is searched with s^{tw} and its gradient. Let $\text{Grad}(s^{tw})$ be the gradient of s^{tw}, $\mathbf{br}_i$ should meet the conditions that the standard deviation of s^{tw}_{br} is smaller than the standard deviation of s^{tw}, and the standard deviation of $\text{Grad}(s^{tw}_{\text{br}})$ is smaller than the standard deviation of $\text{Grad}(s^{tw})$. Besides, $\mathbf{br}_i$ searching operation is finished by a band widow with size $w \times 1$.
   c) Calculating the feature curve of land objects $s^{l'}$, $s^{l'} = \max(\mathbf{I}(\mathbf{M}_L == 1, \boldsymbol{b}_1) - \mathbf{I}(\mathbf{M}_L == 1, :)$. The difference spectrums of all land region pixels in band $\boldsymbol{b}_1$ and other bands are calculated as $\mathbf{I}(\mathbf{M}_L == 1, \boldsymbol{b}_1) - \mathbf{I}(\mathbf{M}_L == 1, :)$. $s^{l'}$ is the maximum value of these difference spectrums in each band, which is used to find feature band subsequently.
   d) The feature curve $s^{l'}$ and $\boldsymbol{b}_2$ is combined to find an image band $\boldsymbol{b}_3$ which can suppress land objects. Assuming that $\boldsymbol{b}_2$ exists in band range $\mathbf{br}_j$, $\boldsymbol{b}_3$ is determined as the minimum value band of $s^{l'}_{\boldsymbol{b}_1}$ - $s^{l'}$ in $\mathbf{br}_j$. More importantly, if the stable band ranges $\mathbf{br}$ do not exist or $\boldsymbol{b}_2$ does not belong to any $\mathbf{br}_i$. The $\boldsymbol{b}_3$ is equal to $\boldsymbol{b}_2$.

The motivation of the above BS steps is to reduce the difference in land objects in the selected bands on the premise of ensuring a significant difference between the underwater target and water. Band $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ can only reflect the difference between the target and water. In the band range $\mathbf{br}_j$ where $\boldsymbol{b}_2$ is located, the difference between the target and water is similar. But the spectrum of land objects may change violently. It is necessary to ensure the difference between land objects is smaller than the underwater target in feature bands. Therefore, $\boldsymbol{b}_3$ is further searched in $\mathbf{br}_j$ to minimize the difference in land objects in band $\boldsymbol{b}_1$ and $\boldsymbol{b}_3$.

Moreover, although the obtained masks are not completely accurate, using the average value as the representative spectrum in step 1) is sufficient to correctly reflect the spectral changes in the target and background.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:03:13 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DETAILED INFORMATION OF THE HSIS IN THE HNU-UTD DATASET

Data name	Image size	Spatial resolution	Imaging time (year-month-day)	Central location	Sensor	Main target
UTD1	560×610×270	0.024m	2020-09-21	36.09°N, 120.53°E	Nano-Hyperspec	Tetrapods/Cement
UTD2	250×250×270	0.024m	2019-09-24	37.84°N, 119.10°E	Nano-Hyperspec	Plant
UTD3	300×300×177	16.4m	2008-08-14	49.51°N, -124.82°E	AVIRIS	Sandstone
UTD4	500×500×177	17m	2018-01-18	19.81°N, -156.02°E	AVIRIS	Rock
UTD5	230×230×177	17m	2018-01-18	19.69°N, -156.04°E	AVIRIS	Rock
UTD6	400×400×177	7.6m	2018-01-29	20.73°N, -156.46°E	AVIRIS	Silt sediment
UTD7	300×300×177	3.4m	2010-07-07	30.32°N, -87.26°E	AVIRIS	Silt sediment
UTD8	200×200×177	3.4m	2010-07-07	30.38°N, -86.25°E	AVIRIS	Silt sediment

3) *Selecting the Feature Bands of Land Objects:* Similarly, the method in step 2) can be used to detect land objects. This step is used to remove the land region in the detection result subsequently.
4) *Generating the Detection Result of the Underwater Target Through Band Math:* At first, the feature bands $\boldsymbol{b}_1$ and $\boldsymbol{b}_3$ are used to obtain the probability map of the underwater target $\mathbf{P}_t$ by

$$\mathbf{P}_t = \frac{\mathbf{I}_{\boldsymbol{b}_1} - \mathbf{I}_{\boldsymbol{b}_3}}{\mathbf{I}_{\boldsymbol{b}_1} + \mathbf{I}_{\boldsymbol{b}_3}}. \tag{8}$$

Similarly, the probability map of the land objects $\mathbf{P}_l$ can be acquired. Then, $\mathbf{P}_l$ is used to optimize $\mathbf{P}_t$ to get the original detection result $\mathbf{P}$ as follows:

$$\mathbf{P} = \begin{cases} 0, & \text{if } \mathbf{P}_l > \alpha \text{ or } \mathbf{P}_t < 0 \\ \mathbf{P}_t, & \text{otherwise} \end{cases} \tag{9}$$

where α is an adaptive threshold which is set as the max value of water region pixels in $\mathbf{P}_l$. Finally, the guided filter [46] is then adopted to reduce noise interference in the original detection result $\mathbf{P}$. The guided image is set as the first three components of the dimension reduced HSI obtained by principal component analysis. To easily understand the process of underwater target feature BS, the detailed steps are summarized in Algorithm 1.

Algorithm 1 Feature BS Procedure of Target

Input: Input HSI $\mathbf{I}$, land mask $\mathbf{M}_L$, target mask $\mathbf{M}_T$, and water mask $\mathbf{M}_{PW}$
Output: Feature bands $\boldsymbol{b}_1$ and $\boldsymbol{b}_3$
1: Calculate the mean spectrum of target s^t and pure water s^w
2: Calculate the difference spectrum of target and pure water, $s^{tw} = s^t - s^w$
3: Find the maximum band $\boldsymbol{b}_1$ and minimum band $\boldsymbol{b}_2$ of s^{tw}
4: Search the stable band ranges of s^{tw}, $\boldsymbol{br}$
5: Find the band range where $\boldsymbol{b}_2$ is located, $\boldsymbol{br}_j$
6: Calculate the max feature curve of land object $s^{l'}$
7: Find the minimum band of $s^{l'}_{\boldsymbol{b}_1}$ - $s^{l'}$ in $\boldsymbol{br}_j$, $\boldsymbol{b}_3$
8: **return** $\boldsymbol{b}_1$ and $\boldsymbol{b}_3$

III. HNU-UTD DATASET

This section introduces a new benchmark dataset called HNU-UTD built for underwater target detection in this study. The HNU-UTD dataset consists of eight individual HSIs captured by Airborne Visible Infrared Imaging Spectrometer (AVIRIS) and Headwall Nano-Hyperspec sensor. Table I lists the detailed information of each data. Fig. 2 displays the true-color representation of these data.

UTD1 and UTD2 were obtained by the Headwall Nano-Hyperspec equipped with Da Jiang Innovations (DJI) Matrice 600 Pro UAV platform. The raw data with push broom imaging mode have 270 bands from 400 to 1000 nm. Besides, the flight height was 40 m. The necessary preprocessing for UTD1 and UTD2 includes radiometric calibration and geometric correction which were performed with the HyperSpec software. The radiometric calibration converts raw digital values into radiance values based on the calibration parameters of the imaging sensor. The geometric correction depends on a collinear equation and the geometric information which can be recorded by the inertial measurement unit modules and global positioning system. Moreover, the ground-truth maps of these two data were obtained through field investigation.

UTD3–UTD8 were acquired via AVIRIS and downloaded from the website.[2] The corresponding ground-truth maps were obtained through visual interpretation based on Google Earth images. AVIRIS is a whisk broom imaging spectrometer which can produce 677 pixels for the 224 detectors in each scan. The original AVIRIS data have 224 bands and cover the spectral range from 366 to 2495 nm with a 9.5-nm spectral resolution. Besides, to get the reflectance data, FLAASH atmospheric correction has been executed on the original radiance data with the ENVI 5.3 software. After removing the invalid bands mainly resulted by water absorbing around 1.4 and 1.9 μm (104–123, 150–172, 221–224), only 177 bands were kept for subsequent processing.

IV. EXPERIMENTS

In this section, experiments are carried out on the HNU-UTD dataset to verify the performance of different detection methods. First, the experimental setting is introduced briefly. Then, the parameter of band window length w in the proposed FBUD method is analyzed. Finally, the detection results of the studied methods in the HUTD task are compared and analyzed.

A. Experimental Setting

1) Evaluation Metrics: The most commonly used metrics in the detection task, the receiver operating characteristic (ROC) curve and the area under the curve (AUC) value, are figured out to measure the efficiency of testing methods quantitatively.

[2]https://aviris.jpl.nasa.gov/dataportal/

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:03:13 UTC from IEEE Xplore. Restrictions apply.

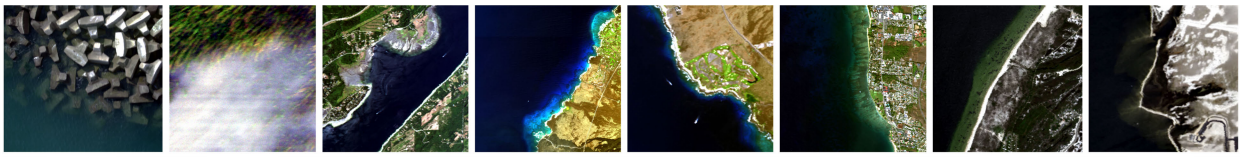


Fig. 2. True color images of the HNU-UTD dataset. (a) UTD1. (b) UTD2. (c) UTD3. (d) UTD4. (e) UTD5. (f) UTD6. (g) UTD7. (h) UTD8.

ROC curve describes the correlation of true positive rate (TPR) and false positive rate (FPR) as the binary threshold changes on the probability map of target. FPR and TPR correspond to the X- and Y-axes in the coordinate plane, respectively. In general, when the ROC curve is closer to the top left of the coordinate plane, the detection performance is better.

The AUC value ranges from 0 to 1 which is the area under the ROC curve, which can quantify the performance indicated by the curve. Thus, AUC is more convenient for performance comparison. Accordingly, the AUC value is higher when the detection result is better. Furthermore, the separability map represented by the Box–and-Whisker plot is used to attain visual analysis of target and background separability.

2) Comparison Methods: In this work, six representative HSI target detection methods and two classification methods are adopted for comparison. HSI target detection methods include CEM [9], hCEM [14], MF [7], ACE [6], HUD [47], and OSP [8]. As mentioned before, CEM is designed as a finite impulse response filter that can minimize the energy of the background. hCEM adds hierarchical filtering processing based on CEM. MF uses the ratio of the conditional probability density functions to determine whether the pixel is like the target signal or not. ACE uses a generalized likelihood ratio approach to get the statistic and achieve target background separation. HUD algorithm uses a multivariate normal distribution to model the background and replaces the estimated abundance with their whitened fully constrained least squares counterparts. OSP develops an orthogonal subspace projector to exclude the response of background. Besides, the two classification methods are Fast3DCNN [48] and MCNN-CP [49]. Fast3DCNN uses a 3-D kernel function over multiple bands to improve the computation efficiency and detection accuracy. MCNN-CP uses a mixed CNN with covariance pooling to acquire feature maps for detection. Since all the studied methods are supervised, a fixed 3×3 pixel window is artificially assigned in the center region of the underwater target area to get the prior target spectrum for six target detection methods. Besides, ten target pixels and ten background pixels are randomly selected as the training samples for Fast3DCNN and MCNN-CP.

3) Parameter Settings: Parameters are set to default values for all the studied methods. For Fast3DCNN and MCNN-CP, the model of final epoch is used to find the target. In the proposed method, the window length w for stable band range searching is needed to be determined. The following experiment will illustrate the basis for setting w.

B. Parameter Analysis

In this section, three HSIs captured from different sensors (UTD1, UTD3, and UTD4) are used to analyze the effect of window length w on the result. The variation interval of parameter w is set to be [3] and [29]. Fig. 3 shows the relationship between AUC value and w. Obviously, the AUC of UTD1 reaches the peak value when $w = 7$. UTD3 gets the minimum AUC value when $w = 9$. However, the AUC values are similar when $w \in [7, 11]$ for UTD4, which means the influence of w on UTD4 is relatively small. Thus, w is set to 7 in the proposed method. The following experiments also illustrate that this default setting is able to obtain satisfactory results.

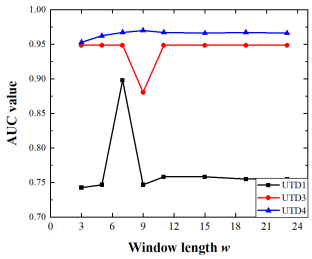


Fig. 3. Influence of the window length w on the detection performance of the proposed method.

C. Underwater Target Detection Performance of Different Methods

The detection results of various methods on the HNU-UTD dataset are presented in Fig. 4. The pixel value in the detection maps represents the probability that the pixel is the target. According to the visual inspection of Fig. 4, FBUD can identify more correct target pixels in UTD2 and UTD8. In the detection maps of FBUD, the target of interest is more obvious compared with the HSI target detection methods. But some water pixels also have high value in the result of UTD3, because FBUD only eliminates the interference of land objects during the feature BS process and does not take the diversity of water into consideration sufficiently. In general, the results of FBUD are more similar to the ground-truth map. For other target detection methods, only the part area of the target can be highlighted in the results due to the variation of spectrum in real-world scenes. Especially in UTD2, they can hardly detect underwater plants.

Overall, these comparative methods achieve greater visual difference with reference maps, where hCEM and HUD obtain better detection maps relatively. CEM and MF have similar performance and detect fewer target pixels. Moreover, hCEM outperforms CEM, which means filtering is beneficial for detection. In the target probability map of ACE, target pixels have the lowest probability values, which indicates ACE is difficult to identify the underwater target. The results of HUD lost some target pixels, especially in UTD2. OSP detects some

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:03:13 UTC from IEEE Xplore. Restrictions apply.

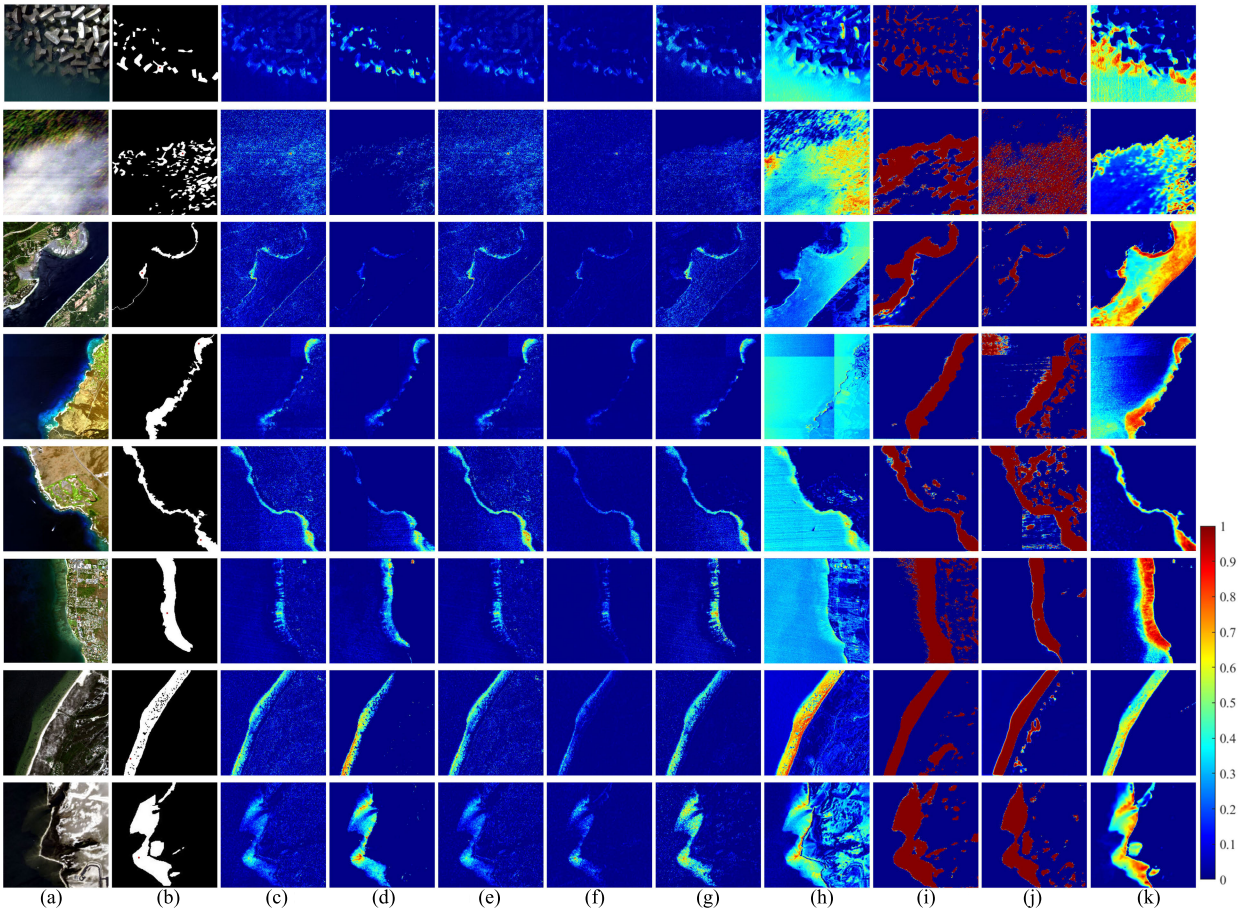


Fig. 4. Underwater target detection results of the studied methods for the HNU-UTD dataset. The red regions in (b) denote the pixels used to provide prior target spectrum for target detection methods. (a) HNU-UTD. (b) Ground-truth map. (c) CEM. (d) hCEM. (e) MF. (f) ACE. (g) HUD. (h) OSP. (i) Fast3DCNN. (j) MCNN-CP. (k) FBUD.

TABLE II
AUC VALUES OF DIFFERENT METHODS ON THE HNU-UTD DATASET

Data name	CEM [9]	hCEM [14]	MF [7]	ACE [6]	HUD [47]	OSP [8]	Fast3DCNN [48]	MCNN-CP [49]	FBUD
UTD1	0.8563	0.8864	0.8475	0.7941	0.8683	0.8206	0.9418	**0.9563**	0.9188
UTD2	0.6606	0.6095	0.6503	0.5439	0.6839	0.6890	0.8233	0.7130	**0.8935**
UTD3	0.8616	0.8367	0.8681	0.8400	0.8839	0.8372	0.9113	0.7796	**0.9530**
UTD4	0.7765	0.8533	0.7699	0.7230	0.8038	0.5462	0.9682	0.7935	**0.9767**
UTD5	0.7848	0.7860	0.7895	0.7957	0.8108	0.8351	0.9624	0.9382	**0.9949**
UTD6	0.6679	0.8842	0.6715	0.7557	0.7203	0.8133	0.9065	0.9607	**0.9971**
UTD7	0.7877	0.8212	0.7920	0.8205	0.8173	0.9906	0.9557	0.9753	**0.9963**
UTD8	0.8303	0.8997	0.8295	0.8264	0.8615	0.8756	0.9557	0.9233	**0.9838**

The bold entries represent the best performance in each row.

wrong target pixels in UTD1 and UTD3–UTD6, but obtains a better result in UTD2. However, the chessboard phenomenon resulting from image quality is considered as a feature and kept in the feature extracting process of OSP. MCNN-CP is also affected by the image quality, and noise pixels exist in the results of UTD4 and UTD6. In addition, since Fast3DCNN and MCNN-CP consider detection as a binary classification problem, the probability value of most target pixels in their results is close to 1. For the two classification methods, the target is easy to distinguish. But some background pixels also have high value in the detection result, especially in UTD2 and UTD5.

AUC values and ROC curves are reported in Table II and Fig. 5, which are complied with the above observation. Based on the evaluation indexes, FBUD obtains the maximum AUC values and its ROC curves are closer to the left top of the coordinate plane in most scenarios. hCEM and HUD achieve better performance than other target detection methods generally. Through Fast3DCNN and MCNN-CP misclassified some background pixels, they find more high-confidence target pixels and obtain higher accuracy in UTD1.

To compare the separability between the underwater target and the background, the box plots of different methods are presented in Fig. 6. The orange and blue boxes represent the probability area of the target and background, respectively. The distance between the orange box and the blue box indicates the separability of the target and the background. When the target and the background are better separated, the distance between the two boxes is larger. In these box plots, abnormal points are ignored to better compare the different methods. According

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:03:13 UTC from IEEE Xplore. Restrictions apply.

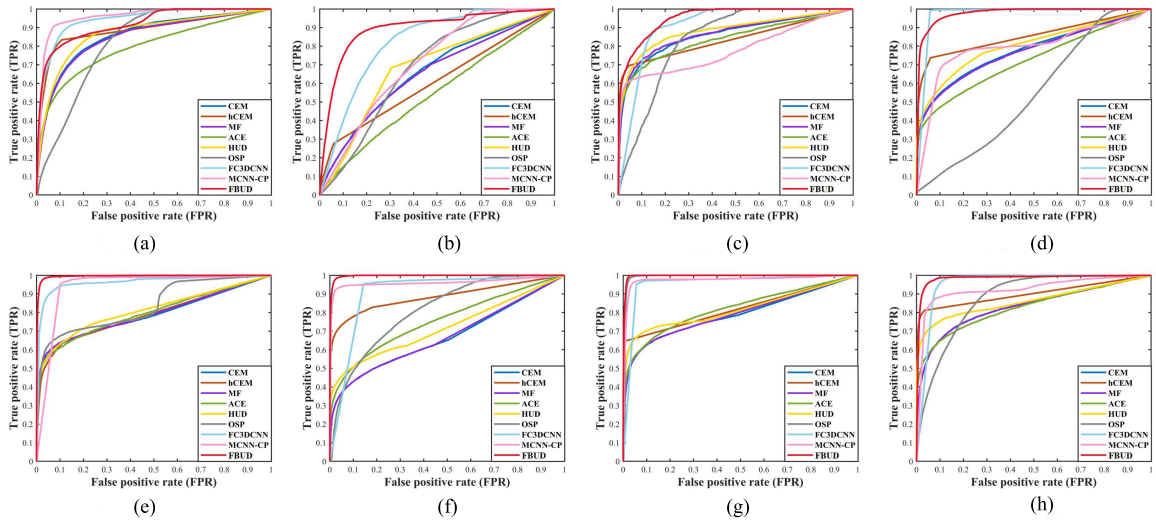


Fig. 5. ROC curves of HNU-UTD dataset obtained by all the studied methods. (a) UTD1. (b) UTD2. (c) UTD3. (d) UTD4. (e) UTD5. (f) UTD6. (g) UTD7. (h) UTD8.

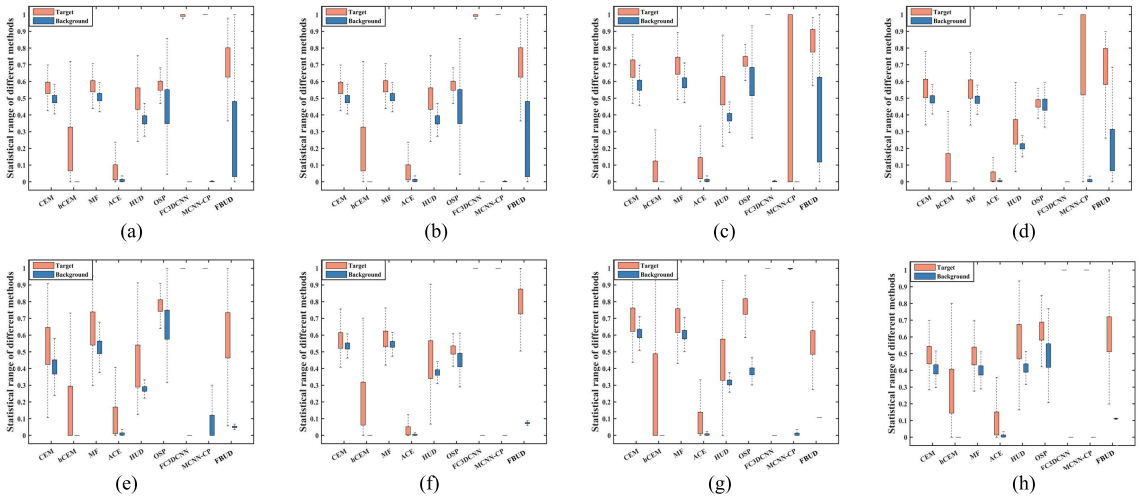


Fig. 6. Statistical separability analysis of the target and background with different methods. (a) UTD1. (b) UTD2. (c) UTD3. (d) UTD4. (e) UTD5. (f) UTD6. (g) UTD7. (h) UTD8.

TABLE III
COMPUTING TIME OF THE STUDIED METHODS (IN SECONDS)

Data	CEM	hCEM	MF	ACE	HUD	OSP	Fast3DCNN	MCNN-CP	FBUD
UTD1 (560×610×270)	**0.45**	5.51	3.22	4.87	29.13	1.95	train:2.94, test:68.19	train:32.32, test:435.42	20.92
UTD3 (300×300×177)	**0.04**	0.33	0.28	0.39	8.15	0.16	train:3.09, test:18.78	train:30.30, test:117.58	4.52
UTD4 (500×500×177)	**0.18**	1.41	1.30	1.95	25.01	0.82	train:3.05, test:50.94	train:30.27, test:332.08	9.09

The bold entries represent the best performance in each row.

to Fig. 6(a)–(h), the proposed FBUD algorithm can separate the target from the background for all the images in the HNU-UTD dataset, which further proves the superior detection performance of the proposed FBUD method. In the results of hCEM, the probability values of most background pixels are zero and the abnormal points are ignored. Thus, only a line that represents the median value of the background pixels exists in the blue boxes' location of hCEM, which means hCEM performs well on background suppression. Similarly, Fast3DCNN and MCNN-CP also work well on background suppression. They obtained the maximum distances between the target and the background, except UTD3.

Finally, Table III lists the time consumption of all the methods on three images with different sizes. Obviously, CEM takes the shortest time. All the traditional detection methods have less computational complexity on the whole. Besides, the computational efficiency of Fast3DCNN is superior to MCNN-CP, but they cost a long time for testing the image per patch. The computational efficiency of FBUD is relatively higher since the unmixing process needs some time. Overall, the calculation time of FBUD is acceptable. Moreover, it should be noted that the experiments are implemented on the 3.7-GHz CPU and NVIDIA GeForce GTX 1080 Ti GPU computer.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:03:13 UTC from IEEE Xplore. Restrictions apply.

According to the above analysis, the FBUD method has satisfactory performance both in subjective and objective evaluation. But noise is still present in detection maps of UTD1 and UTD3, since the spectral information is mainly used for detection. Other supervised methods (e.g., CEM, MF, and OSP) also have the same problem. The noise problem can be further considered in the subsequent studies. In this work, only single-class target detection is considered. However, detecting underwater targets of multiple classes is also an interesting research topic. For example, unsupervised unmixing can be used for underwater targets. The distribution of underwater targets can also be obtained through duplicate detection.

V. Conclusion

In this work, a novel feature-band-based unsupervised method for HUTD near the coastline is proposed, which fully uses spectral characteristics. First, NDWI is adopted to obtain a coarse water region and land mask. Then, spectral unmixing is conducted on the water region to obtain the abundance map of the target. Through binarizing the abundance map, desired target mask is generated. According to these masks, representative spectrums of land, pure water, and target can be calculated. Next, the difference spectrum is used to guide the selection of feature bands. Finally, the feature bands are combined to attain the probability map of underwater target. Experiments performed on the constructed HNU-UTD dataset demonstrate that FBUD is superior to many supervised methods.

This work explores the intrinsic spectral feature of underwater targets and does not fully take the spatial information into account, which can be further improved. Furthermore, we also find that the unmixing result can provide a false label for detection, which may be helpful for supervised detectors.

References

[1] C.-I. Chang, "Hyperspectral anomaly detection: A dual theory of hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511720.

[2] F. A. Kruse, J. W. Boardman, and J. F. Huntington, "Comparison of airborne hyperspectral data and EO-1 Hyperion for mineral mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1388–1400, Jun. 2003.

[3] B. Du, Y. Zhang, L. Zhang, and D. Tao, "Beyond the sparsity-based target detector: A hybrid sparsity and statistics-based detector for hyperspectral images," *IEEE Trans. Image Process.*, vol. 25, no. 11, pp. 5345–5357, Nov. 2016.

[4] W. Xie, J. Lei, S. Fang, Y. Li, X. Jia, and M. Li, "Dual feature extraction network for hyperspectral image analysis," *Pattern Recognit.*, vol. 118, Oct. 2021, Art. no. 107992.

[5] T. Wang, B. Du, and L. Zhang, "An automatic robust iteratively reweighted unstructured detector for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2367–2382, Jun. 2014.

[6] S. Kraut, L. L. Scharf, and R. W. Butler, "The adaptive coherence estimator: A uniformly most-powerful-invariant adaptive detection statistic," *IEEE Trans. Signal Process.*, vol. 53, no. 2, pp. 427–438, Feb. 2005.

[7] F. C. Robey, D. R. Fuhrmann, E. J. Kelly, and R. Nitzberg, "A CFAR adaptive matched filter detector," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 28, no. 1, pp. 208–216, Jan. 1992.

[8] J. C. Harsanyi and C.-I. Chang, "Hyperspectral image classification and dimensionality reduction: An orthogonal subspace projection approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 4, pp. 779–785, Jul. 1994.

[9] W. H. Farrand and J. C. Harsanyi, "Mapping the distribution of mine tailings in the Coeur d'Alene River Valley, Idaho, through the use of a constrained energy minimization technique," *Remote Sens. Environ.*, vol. 59, no. 1, pp. 64–76, Jan. 1997.

[10] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Sparse representation for target detection in hyperspectral imagery," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 629–640, Jun. 2011.

[11] Y. Zhang, B. Du, and L. Zhang, "A sparse representation-based binary hypothesis model for target detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1346–1354, Mar. 2015.

[12] A. W. Bitar, L.-F. Cheong, and J.-P. Ovarlez, "Sparse and low-rank matrix decomposition for automatic target detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5239–5251, Aug. 2019.

[13] X. Yang, J. Chen, and Z. He, "Sparse-SpatialCEM for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2184–2195, Jul. 2019.

[14] Z. Zou and Z. Shi, "Hierarchical suppression method for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 330–342, Jan. 2015.

[15] X. Jiao and C.-I. Chang, "Kernel-based constrained energy minimization," in *Proc. 15th SPIE Algor. Technol. Multispectral Hyperspectral Ultraspectral Imagery*, vol. 6966, 2008, pp. 69661S1–69661S11.

[16] X. Yang, M. Zhao, T. Gao, J. Chen, and J. Zhang, "Multiscale-superpixel-based sparsecem for hyperspectral target detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2021.

[17] W. Li, Q. Du, and B. Zhang, "Combined sparse and collaborative representation for hyperspectral target detection," *Pattern Recognit.*, vol. 48, no. 12, pp. 3904–3916, Dec. 2015.

[18] M. Imani, "Attribute profile based target detection using collaborative and sparse representation," *Neurocomputing*, vol. 313, pp. 364–376, Nov. 2018.

[19] Y. M. Zhang, B. Du, Y. Zhang, and L. Zhang, "Spatially adaptive sparse representation for target detection in hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 1923–1927, Nov. 2017.

[20] W. Xie, J. Yang, J. Lei, Y. Li, Q. Du, and G. He, "SRUN: Spectral regularized unsupervised networks for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1463–1474, Oct. 2020.

[21] Y. Li, Y. Shi, K. Wang, B. Xi, J. Li, and P. Gamba, "Target detection with unconstrained linear mixture model and hierarchical denoising autoencoder in hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 31, pp. 1418–1432, 2022.

[22] Y. Shi, J. Li, Y. Zheng, B. Xi, and Y. Li, "Hyperspectral target detection with RoI feature transformation and multiscale spectral attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5071–5084, Jun. 2021.

[23] W. Xie, J. Zhang, J. Lei, Y. Li, and X. Jia, "Self-spectral learning with GAN based spectral–spatial target detection for hyperspectral image," *Neural Netw.*, vol. 142, pp. 375–387, Oct. 2021.

[24] G. Zhang, S. Zhao, W. Li, Q. Du, Q. Ran, and R. Tao, "HTD-Net: A deep convolutional neural network for target detection in hyperspectral imagery," *Remote Sens.*, vol. 12, no. 9, p. 1489, 2020.

[25] D. Zhu, B. Du, and L. Zhang, "Two-stream convolutional networks for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6907–6921, Aug. 2021.

[26] D. Manolakis, C. Siracusa, and G. Shaw, "Hyperspectral subpixel target detection using the linear mixing model," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 7, pp. 1392–1409, Jul. 2001.

[27] X. Yang, M. Dong, Z. Wang, L. Gao, L. Zhang, and J.-H. Xue, "Data-augmented matched subspace detector for hyperspectral subpixel target detection," *Pattern Recognit.*, vol. 106, Oct. 2020, Art. no. 107464.

[28] Z. Huang, L. Fang, and S. Li, "Subpixel-pixel-superpixel guided fusion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 5998–6007, Sep. 2020.

[29] W. Rao et al., "Siamese transformer network for hyperspectral image target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526419.

[30] F. Vincent and O. Besson, "One-step generalized likelihood ratio test for subpixel target detection in hyperspectral imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4479–4489, Jun. 2020.

[31] S. Jay and M. Guillaume, "Underwater target detection with hyperspectral remote-sensing imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2010, pp. 2820–2823.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:03:13 UTC from IEEE Xplore. Restrictions apply.

[32] S. Jay, M. Guillaume, and J. Blanc-Talon, "Underwater target detection with hyperspectral data: Solutions for both known and unknown water quality," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 4, pp. 1213–1221, Aug. 2012.
[33] D. B. Gillis, "An underwater target detection framework for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1798–1810, 2020.
[34] Z. Xia and Y. Gu, "Parameter feature extraction for hyperspectral detection of the shallow underwater target," *Sci. China Technol. Sci.*, vol. 64, no. 5, pp. 1092–1100, Mar. 2021.
[35] J. Qi, Z. Gong, W. Xue, X. Liu, A. Yao, and P. Zhong, "An unmixing-based network for underwater target detection from hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 5470–5487, 2021.
[36] J. Qi et al., "A self-improving framework for joint depth estimation and underwater target detection from hyperspectral imagery," *Remote Sens.*, vol. 13, no. 9, p. 1721, Apr. 2021.
[37] C.-H. Yeh et al., "Lightweight deep neural network for joint learning of underwater object detection and color conversion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6129–6143, Nov. 2022.
[38] M. Pedersen, J. Bruslund Haurum, R. Gade, and T. B. Moeslund, "Detection of marine animals in a new underwater dataset with varying visibility," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, Jun. 2019, pp. 18–26.
[39] W.-H. Lin, J.-X. Zhong, S. Liu, T. Li, and G. Li, "ROIMIX: Proposal-fusion among multiple images for underwater object detection," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 2588–2592.
[40] X. Fu, X. Shang, X. Sun, H. Yu, M. Song, and C.-I. Chang, "Underwater hyperspectral target detection with band selection," *Remote Sens.*, vol. 12, no. 7, p. 1056, Mar. 2020.
[41] J. Qi et al., "Bathymetric-based band selection method for hyperspectral underwater target detection," *Remote Sens.*, vol. 13, no. 19, p. 3798, Sep. 2021.
[42] S. K. McFeeters, "The use of the normalized difference water index (NDWI) in the delineation of open water features," *Int. J. Remote Sens.*, vol. 17, no. 7, pp. 1425–1432, May 1996.
[43] M. E. Winter, "N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data," *Proc. SPIE*, vol. 3753, pp. 266–275, Oct. 1999.
[44] N. Keshava and J. F. Mustard, "Spectral unmixing," *IEEE signal Process. Mag.*, vol. 19, no. 1, pp. 44–57, Aug. 2002.
[45] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-9, no. 1, pp. 62–66, Jan. 1979.
[46] K. He, J. Sun, and X. Tang, "Guided image filtering," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 6, pp. 1397–1409, Jun. 2013.
[47] J. Broadwater and R. Chellappa, "Hybrid detectors for subpixel targets," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 29, no. 11, pp. 1891–1903, Nov. 2007.
[48] M. Ahmad, A. M. Khan, M. Mazzara, S. Distefano, M. Ali, and M. S. Sarfraz, "A fast and compact 3-D CNN for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[49] J. Zheng, Y. Feng, C. Bai, and J. Zhang, "Hyperspectral image classification using mixed convolutions and covariance pooling," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 522–534, Jan. 2021.



Shuo Zhang received the B.S. degree from Nanchang Hangkong University, Jiangxi, China, in 2018. She is currently pursuing the Ph.D. degree with the Laboratory of Vision and Image Processing, Hunan University, Changsha, China.

Her research interests include hyperspectral image classification and target detection.



Puhong Duan (Member, IEEE) received the B.S. degree from Suzhou University, Suzhou, China, in 2014, and the Ph.D. degree from Hunan University, Changsha, China, in 2021.

From October 2019 to December 2019, he was a Visiting Ph.D. Student with the Faculty of Electrical Engineering and Computer Science, Technische Universitat Berlin, Berlin, Germany. From January 2020 to October 2020, he was a Visiting Ph.D. Student with the Helmholtz-Zentrum Dresden-Rossendorf, Helmholtz Institute Freiberg for Resource Technology, Freiberg, Germany. He is currently a Post-Doctoral Fellow with the College of Electrical and Information Engineering, Hunan University. His research interests include image restoration, multimodal image fusion, and classification.

Dr. Duan was a Finalist for the Best Student Paper Award at the International Geoscience and Remote Sensing Symposium (IGARSS) 2020.



Xudong Kang (Senior Member, IEEE) received the B.S. degree from Northeastern University, Shenyang, China, in 2007, and the Ph.D. degree from Hunan University, Changsha, China, in 2015.

In 2015, he joined the College of Electrical Engineering, Hunan University, where he is currently a Full Professor with the School of Robotics. He has authored or coauthored more than 100 refereed articles. His research interests include hyperspectral image processing, multimodal image fusion, and advance artificial intelligence methods.

Dr. Kang received the National Nature Science Award of China in 2019. He was selected as the Best Reviewer for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS and the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He was an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING from 2018 to 2019. He serves as an Associate Editor for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS and the IEEE JOURNAL ON MINIATURIZATION FOR AIR AND SPACE SYSTEMS.



Yan Mo received the B.S. degree from the Changchun University of Technology, Changchun, China, in 2015, the M.S. degree from Xiangtan University, Xiangtan, China, in 2018, and the Ph.D. degree from Hunan University, Changsha, China, in 2022.

Since 2022, he has been with the School of Electrical and Information Engineering, Changsha University of Science and Technology, Changsha. His research interests include multisource image matching and image fusion.



Shutao Li (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees from Hunan University, Changsha, China, in 1995, 1997, and 2001, respectively, all in electrical engineering.

In 2001, he joined the College of Electrical and Information Engineering, Hunan University. From May 2001 to October 2001, he was a Research Associate with the Department of Computer Science, The Hong Kong University of Science and Technology, Hong Kong. From November 2002 to November 2003, he was a Post-Doctoral Fellow with the Royal Holloway College, University of London, London, U.K. From April 2005 to June 2005, he was a Visiting Professor with the Department of Computer Science, The Hong Kong University of Science and Technology. He is currently a Full Professor with the College of Electrical and Information Engineering, Hunan University. He has authored or coauthored more than 200 refereed articles. His research interests include image processing, pattern recognition, and artificial intelligence.

Dr. Li received two Second-Grade State Scientific and Technological Progress Awards of China in 2004 and 2006. He is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and the IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT. He is a member of the Editorial Board of the *Information Fusion* and the *Sensing and Imaging*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:03:13 UTC from IEEE Xplore. Restrictions apply.